

Abhay Gupta

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Portfolio: [abgup.com](https://notes.abgup.com) & notes.abgup.com

Current: **Seattle, WA**

U.S. Citizen

PUBLICATIONS, CONFERENCES, AND PATENTS

Resume coded in LaTeX

Automobile Damage Detection Using Thermal Conductivity, pdf

Filed: Dec 2018

- Inventors: J. Schow, and **A. Gupta**
- **U.S. Patent** #: 11,580,791

Feb 2023

A Cellular Automaton for Modeling Non-Trivial Biomembrane Ruptures, pdf

Submitted: Jul 2018

- **A. Gupta**, G. Reint, I. Gozen, and M. Taylor
- Presented at **The 13th World Congress in Computational Mechanics (WCCM)**
- Session: *Novel Mathematical Models and Computational Methods*, New York, NY
- Published in **Soft Matter**

Sep 2018

Apr 2019

SKILLS

Languages	Frameworks	Software Tools	Modelling/Analysis	Hardware Tools
• Python & Lua • C# & Java • Javascript & VBA • HTML/CSS • Bash & Powershell • Vimscript • C/C++ • LabView	• Django • WinForms/WPF • Flask • Jekyll • Cordova • JQuery • Next.js • LaTeX & Maple • Matlab/Simulink	• Git • Vim/Neovim • Cmake • Jira/Azure • Docker • Figma • Microsoft Office • Linux • Android/iOS	• SolidWorks • Siemens NX • OnShape/Auto-CAD • Ansys CFX • Star CCM+ • Abaqus • FreeCAD • Fusion 360	• 3D Printers • Lathe/Mill • Drill Press • Bandsaws • Sanders/Grinders • Oscilloscopes • Soldering • Function Generators

EDUCATION

University of Washington, M.S. Robotics & Data Science

2018-2019

- Thesis: 3D Graphics & Regression Models || 3.6
 - Advisors: Steve Brunton & Ashis Banerjee
 - Taught graduate mathematics courses for engineering
 - Researched & developed a custom physics simulation of Optical Tweezers and digital twin models for compliant motors via machine learning regression methods
- Python || MATLAB || C++ OpenGL || ROS

Santa Clara University, M.S. Computer Engineering

2017-2018

- Transferred to University of Washington || 3.8
 - Simulated biological membrane fracture through AI methods (see paper in publications)
- C || C++

Santa Clara University, B.S. Mechanical Engineering

2014-2017

- Entrepreneurship minor || Graduated in 3 years with honors || 3.6
 - Assisted teaching course in Numerical Analysis to undergraduate engineering students
- MATLAB || LaTeX || Maple Simulink || LabVIEW SolidWorks || Abaqus || Star-CCM+ Oscilloscopes || Lathe || Mill

INDUSTRY EXPERIENCE

Pebble Products LLC, Founder

2023 - Present

- Develop/prototype new hardware/software products
 - Contract engineering services to local companies/individuals
- notes.abgup.com pebbleproducts.com

Kawaskai Robotics, Software Engineer, Robotics

2022

- Developed software to move wafer handling (scara) robots
 - Tested physical robotics arms on local and vendor facilities to ensure functionality
 - Optimized for throughput and reach requirements to maximize computer chip production capability
- C language AS (DSL)

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SummerBio, Software Engineer, Robotics	2021 - 2022
- Developed software drivers (6 DOF arms, benchtop systems) for VWorks - Built communication platforms through both Ethernet and serial port protocols - Built hardware testing rigs to ensure new hardware/drivers work with automation line - Unit tested drivers through manual isolation testing and automated via C# test framework, xUnit	
C# Python VBA	Figma
Lam Research, Software & Mechanical Engineer	2020 - 2021
- Led and assisted Android app development via Cordova (JS) framework - Developed/Maintained software to optimize a 6 DOF robotic arm via C# WinForms - Built communication protocols for hardware components via Modbus, Ethernet, and Bluetooth protocols - Physically tested software and hardware in clean room environment via systematic approach - Analyzed test results via data science techniques to ensure repeatability and reliability metrics - Led development of and maintained a fishbone diagram issue diagnosis application with offshore developers via C# Xiamarin Framework on Windows and Android - Automated Android content upload and verification via VBA excel automation tools - Worked alongside machinists to develop custom test rigs and hardware components (sheet metal, aluminum/steel parts, injection molded designs, etc) for robotics arm applications - Designed and printed custom 3D components on a MakerBot and local vendors - Designed a custom electronics cart via Siemens NX fitting custom components and facility constraints	
C# JavaScript Python	NX Teamcenter
Figma	VBA
CSAA Insurance, Physics Consultant	2018 - 2020
- Evaluated novel engineering and physics aspects for patent applications - Researched alternative approaches for products and methods	

INTERNSHIPS

Microvision, SDE Intern	Summer 2019
Modeled the response of Lidar activated SiPM (Silicon photomultipliers)	
Simulink MATLAB	LTSpice
TheraNova, SDE Intern	Summer 2018
- Developed a python software analysis system to understand gait measurements - Ensured the software analysis is accurate for 80+ patients	
Python MATLAB	
Valeo, Systems Engr Intern	Spring 2018
- Produced hardware and software demos for automotive OEMs - Collaborated with start-ups and OEMs to develop new cabin safety features	
VBA	
Pentair, ME/EE Intern	Fall/Winter 2017
- Optimized performance of steady state and transient phases of circuit breakers - Laboratory tested and simulated rail heating to melt snow	
Ansys CFX SolidWorks	Oscilloscopes, Function Generators
DC/AC Power Supplies	Soldering
Accel Biotech, ME Intern	Summer 2016
- Prototyped medical device components and test assemblies - Supported mechanical, electrical, and software design of a blood diagnostic device	
SolidWorks, 3D Printing	Oscilloscopes
Caltrans, ME Intern	Summer 2015
- Reviewed and advised on structural testing for next generation locomotives - Designed a floor plan using Microsoft Visio & participated in vendor meetings	
Microsoft Visio	

CERTIFICATIONS

Engineering-In-Training, State of CA	May 2018
Solidworks CSWA, Dassault Systèmes	Mar 2015

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VOLUNTEERING

Auto Angels, Bellevue, Car Mechanic	Jun 2024 - Present
Rainier Scholars, Seattle, Computer Science Teacher	Jun 2024 - Present
The Bikery, Seattle, Bicycle Board Member & Technician	Nov 2023 - Present
Capitol Hill Tool Library, Seattle, Volunteer & 3D Print Assistant	May 2024 - May 2025
Santa Clara University, Alumni, Engineering Mentor	Oct 2023 - Jun 2025
FIRST Robotics Competition (FRC) 254, Engineering Mentor	Aug 2021 - Present
Reddit Group: r/ControlTheory, Moderator	Dec 2018 - Present
San Jose Bicycle Clinic, Bicycle Technician	Jul 2020 - Jan 2022

SOFTWARE PROJECTS

3D Modellings via Python, **Developer/Fabricator**

- Create 3d models via Python, then 3d print models to sell/use/gift
Python || Bash github.com/cad-scripts build123d.com

206bikepolo.com, **Web Developer**

- Update and maintain website using Javascript framework (Next.js)
HTML || CSS || Javascript [Next.js](https://next.js.org) || [React](https://react.dev) 206bikepolo.com

SheldonBrown.com, **Web Developer**

- Build and maintain website to share bicycle repair information
- The most popular bike repair website worldwide
HTML || CSS || Javascript [Wordpress](https://wordpress.org) sheldonbrown.com

Find a Paint, **Web Developer**

- Developed website to compare paint brands to help high school mom find corresponding paint for work
Python || HTML/CSS || JS [Flask](https://flask.palletproject.org) findapaint.com

Masala Blend, **Web Developer**

- Help stepmother build webpage to sell homemade spices locally
HTML || CSS || Javascript [Shopify](https://shopify.com) masalablend.com

Personal Website & Resume, **Web Developer**

- Build personal website via github pages & programmatically render resume via Latex
HTML || CSS || Javascript [Jekyll](https://jekyllrb.com) [LaTeX](https://www.latexlive.net) || [LuaLaTeX](https://lualatex.org) abgup.com

Peer Portfolio Consulting, **Web Developer**

- Assist undergraduates build their resume & web portfolio to gain industry experience
HTML || CSS || Javascript [Python](https://python.org) || [Latex](https://www.latexlive.net) [Django](https://djangoproject.com) github.com/carly85/CV

HARDWARE PROJECTS

Speaker Amplifier, **Engineer/Fabricator**

2022-Present

- Designed/built custom speaker amplifier to be compatible with passive speakers
[Soldering](https://www.soldering.com) [BandSaw/Drill Press](https://www.band-saw.com) [KiCAD](https://www.kicad.org) notes.abgup.com/amp

Bike Polo, **Mallet Design/Fabrication**

- Model, 3D print, and fabricate custom bike polo mallet
- Iterate through design process, optimizing for weight and reduction of stress concentration/shattering
[Onshape](https://onshape.com) || [BambuLabs](https://bambu.com) [Drill Press](https://www.drillpress.com) || [Bandsaw](https://www.bandsaw.com) [Ender 3](https://ender3.com) notes.abgup.com/mallet

LED Hula Hoop, **Design/Fabrication**

- Build custom LED timed lighting to match the circadium rhythm of the sun
- Change lighting within building from blue light to gradually hit red light throughout day
[Arduino](https://www.arduino.cc) || [C/C++](https://en.cppreference.com) [Soldering](https://www.soldering.com) [notes/hula-hoop](https://notes.abgup.com/hula-hoop)

HOBBIES

Bicycling || Handstands || Basketball || Bike Polo || Unicycling || Tennis

Running || Skateboarding || Rollerblading || Dancing

Weight Lifting || Yoga

Board Games || Drawing || Reading || Cooking

Coding || DIY Projects || Electronics || 3D Printing/CAD || Fabrication || Robotics

For more: abgup.com